

- Q.19 Explain thread gauges.
Q.20 Write short note on welding fixture.
Q.21 Explain location for various services.
Q.22 Explain procedure for lathe fixture design.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Design and draw a milling fixture for a dovetail slot of 10mm depth on a cubical block of 100 x 100 x 20mm size. Assume any other dimensions.
Q.24 Explain the procedure of designing a snap plug gauge for a shaft with 40 mm outside diameter. Assume and give proper tolerances.
Q.25 Write short notes on following:
a) Taylor's principle
b) Gang milling fixture

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4th Sem / Mechanical (Tool & Die Design)

Subject : Jigs, Fixtures and Gauges - Design and Drawing

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 A progressive plug gauge has "Go" and "NOT Go" sizes on the _____ end.
a) Different
b) Same
c) Only Go end is there
d) Only NOT-GO end is there
Q.2 Vee-type locators is best suitable for location of a _____ shaped job.
a) Square shaped b) Wedge shaped
c) Round shaped d) All of the above

- Q.3 Cost wise Jigs and Fixtures causes the_____.
- Increased Cost of production
 - Less cost in mass production
 - Facilities deployment of less labour
 - All of the above
- Q.4 Which type of jig is used drilling on many directions?
- plate jig
 - Post jig
 - Table jig
 - Box jig
- Q.5 3-2-1 method of location in jigs and fixture, restricts _____ degrees of freedom
- 12
 - 6
 - 9
 - 18
- Q.6 The bushes in Drill jig are hardened to:
- Protect the jig from damage
 - Ensure prolonged life to Less wear and tear.
 - Guide the tool so that it does not become inclined
 - Allow the chips to come out easily

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Jigs are generally used for_____ and_____ operations.
- Q.8 The fixtures are_____tools. (machining/precision)
- Q.9 There are _____ number of degrees in space.
- Q.10 “Go” side of ring gauge has diameter equal to _____ size of the job.
- Q.11 Define thread gauge.
- Q.12 The material general used for Gauges is_____.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain with example the concept of interchangeability.
- Q.14 Explain swing leaf jig and latch jig with diagrams.
- Q.15 Differentiate between jigs and fixture.
- Q.16 Explain and draw any four location devices.
- Q.17 Explain GO and NOT GO gauges.
- Q.18 Write the application of pneumatics in fixtures.